

Proposal for:
John Gilson
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Proposal Date:
2022-08-18

Proposal Number:
QU-0045

Reference
US NY GILSON Saranac

Understanding of project

Gilson Engineering Associates has proposed a ground source heat pump (GSHP) system for the Saranac Lofts project located in Saranac Lake, NY. The project consists of 2 buildings, each with four floors with a total of 70 residential apartments. The main floor of one of the buildings is office space. The total area of the two buildings is approximately 77,000 ft².

Gilson Engineering Associates has requested a proposal to develop an hourly energy model and ground heat exchanger model and design for the project and assistance in selecting heat pump equipment and a system mechanical schematic.

Proposed design methodology

Accurate hourly energy load profiles of the buildings connected to a GHX are critical to the development of a GSHP system that is cost-effective and performs efficiently and predictably for the life of the building. Energy loads are determined by simulating building energy models based on the architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data of the location.

GEOptimize will develop an hourly energy model based on drawings, specifications, and for the project provided by Gilson Engineering Associates. The energy model will be developed in Trane Trace 700 or Trace 3D specifically for the outputs required to import into Ground Loop Design (GLD2016) to model the size and performance of a proposed GHX. Iterations of the energy model *may* be run to determine the impact of several energy efficiency measures on the annual energy load balance and the proposed GHX design. Iterations will include:

- Alternate glass specifications with different U-values and solar heat gain coefficients to modify cooling and heating energy and peak loads.
- Calculation and inclusion of domestic hot water to reduce heat rejection to the GHX
- Exhaust air energy recovery system to reduce overall heating and cooling loads

- Other discretionary heating and / or cooling loads that may become apparent during design development

The information will be used to create a set of iterative hourly energy models that may be used to inform the design with recommendations of changes that can potentially be made to the building and / or mechanical systems that will result in a reduction in the size / cost of the GHX needed to service the building loads.

Thermal properties of the soil and rock at the site will be estimated based on available geological information for the area and / or interviews with drilling contractors familiar with the area. Consideration is given to all potential GHX designs and configurations based on building energy loads, geology, land area available for construction of the GHX and on the capabilities of local GHX contracting firms. Potential GHX designs may include vertical borehole configurations, horizontally excavated or horizontally bored GHX fields as well as groundwater options.

Analysis of building energy and GHX models will often indicate the amount of heat rejected to the GHX is significantly different than the amount of heat extracted. This implies the energy loads to and from the GHX are unbalanced. Using energy modeling to inform the design can significantly improve the energy balance to and from the GHX. This will reduce the size and cost of the GHX while helping ensure long-term efficient performance.

Analysis of GHX models developed based on the information may indicate the amount of energy that must be extracted from the ground to heat the buildings is unequal to the amount of heat that will be rejected to it while cooling the buildings. Additional iterations of the building energy models will be developed to determine the impact of a range of energy efficiency measures on the energy balance and the size and cost of the GHX. Balancing the energy loads:

- Reduces the size of the GHX, reducing first cost
- Avoids long-term temperature increase or decrease in the ground that affects the long-term viability of the system, and affects system efficiency and energy cost
- Creates a system that operates predictably into the future

Scope of work

Energy load profiles

Detailed hourly energy models will be developed using Trane Trace 700 or Trace 3D based on drawings, specifications and information for each building provided by Gilson Engineering Associates. There will be two energy models developed:

- 1) Energy model specifically used to produce a .GL1 file for importing into GLD2016.
- 2) Baseline building energy model for clean heat incentive.

Detailed GHX design

Details of the GHX sketches provided will include (AutoCAD drawings NOT included)

- Layout of GHX circuit piping for the recommended GHX configuration
- Design of fusion welded headers to connect GHX circuits to supply / return runout piping
- Supply / return runout piping from GHX headers to manifold in the building
- Specifications for heat transfer fluid and estimated heat transfer fluid volume
- GHX manifold design up to isolation valves between heat pump system piping and GHX

- Pressure-drop calculations for the GHX and recommended circulation pump sizing and control of the pump
- Recommendations for heat pump selection/sizing
- GHX specifications

Controls

Control strategy for GHX pump and integration with controls for building circulation pump

Deliverables

GEOptimize will prepare a PowerPoint presentation to be delivered to Gilson Engineering Associates and the client at a meeting via screen sharing software detailing the conceptual system design. GEOptimize will also work with Gilson Engineering Associates to develop energy model for incentive and put together report to obtain incentive. The construction drawings will be provided to the engineer of record as sketches.

GEOptimize will work with the engineer of record to ensure a proper design is achieved and that they are confident with the assumed liability of signing off on the system.

Pricing

Description	Rate	Qty	Line Total
Detailed Geo Energy Model	\$16,000.00	1	\$16,000.00
Baseline Building Energy Model	\$4,000.00	1	\$4,000.00
Detailed GHX Design	\$9,000.00	1	\$9,000.00
Proposal Total (USD)			\$29,000.00

Terms

- Unless there are significant changes to the building or scope of work outlined in this proposal, no additional design fees are anticipated. If significant changes are requested, additional work will be charged at \$225/hr.
- Invoices will be submitted monthly for work completed to date. Payment is expected in the first week of January 2023 when payment is made by the client
- GEOptimize carries general liability and professional liability insurance to a maximum of \$2,000,000

Authorized signature – Gilson Engineering Associates